DECK 19 · WEEK 10 · AFTERNOON 3

Alternative Investments

Private capital, real assets, hedge funds, and digital assets — what traditional long-only portfolios cannot do

CFA Level I · Vol 8 Learning Modules 1–7

Roadmap — Alternative Investments

7 sections · ~2 hours · Built bottom-up from CFA readings

01	**Features, methods, structures** 3 alt categories · fund / co-invest / direct · LP-GP term sheet & committed capital
02	**Performance & fees** J-curve · IRR vs MOIC · 2-and-20 with 8% hurdle · catch-up, HWM, clawback
03	**Private capital — PE & private debt** VC / growth / buyout · LBO IRR build · exits · direct lending / mezz / distressed
04	**Real estate & infrastructure** Core → opportunistic · cap rate vs REIT FFO multiple · BOT infra, greenfield / brownfield
05	**Natural resources** Farmland & timberland · commodities via futures · roll yield, contango vs backwardation
06	**Hedge funds** L/S equity · event-driven · relative value · macro / CTA · 2-and-20, HWM, lockups
07	**Digital assets** DLT, PoW vs PoS · crypto vs tokens · direct wallet / futures / ETFs · custody & regulatory risk

Learning objectives

By the end of this deck you should be able to:

1. Classify any alternative investment into private capital, real assets, or hedge funds, and describe the access methods (fund / co-invest / direct) and their respective control-vs-fees trade-offs.

2. Compute a GP's performance fee under 2-and-20 with hard and soft hurdles, with and without a catch-up clause; define high-water mark and clawback.

3. Distinguish IRR from MOIC / DPI / TVPI and explain why IRR is the preferred headline return for long-lived alts.

4. Categorise private equity (VC, growth, buyout) and private debt (direct lending, mezzanine, venture debt, distressed) and place each on the risk–return ladder.

5. Value a piece of commercial real estate using a cap rate on NOI and value a REIT using an FFO multiple, and explain why they can diverge in real time.

6. Describe hedge-fund strategy taxonomy (equity L/S, event-driven, relative value, macro, multi-manager) and the core investment features of digital assets including custody and regulatory risks.

SECTION 01

Features, methods, structures

Alternatives are grouped by what they are NOT — they lack the liquidity, transparency, and easy-to-benchmark returns of public equity and public debt

Three broad categories, five features in common — longer horizon, illiquidity, specialised knowledge, complex fees, appraisal-based valuation.

Category	Sub-classes	Investment approach
Private capital	Private equity (VC / growth / buyout), private debt	Long-horizon ownership or lending to non-public firms
Real assets	Real estate, infrastructure, natural resources	Tangible physical ownership – leases, tolls, harvests
Hedge funds	L/S equity, event-driven, relative value, macro	Leverage + shorting + derivatives on traditional assets

FIVE FEATURES IN COMMON

1. Specialised knowledge to value cash flows and risks. 2. Low correlation with traditional asset returns — the headline diversification pitch. 3. Illiquidity, long horizons, large capital outlays. 4. Performance-based, not flat, compensation to align GP incentives. 5. Appraisal-based valuations (Level 3 fair value) — risk of smoothed, understated volatility.

Source: CFA Institute, Curriculum Vol 8, Learning Module 1, pp. 6–11.

Fund investing, co-investing, and direct investing trade off control for access on a sliding scale of sophistication

Most institutional investors start in funds, graduate to co-investments, and — if large enough — run direct investing in-house (sovereign wealth funds, largest pensions).

FUND INVESTING

Commit capital to a GP; GP sources, selects, and manages the underlying deals. No control over individual investments.

- Typical fee: 1.5 – 2.0% mgmt + 20% carry
- Control: lowest
- Suited for: LPs without specialised in-house team

CO-INVESTING

Invest in the fund AND have rights to deploy additional capital directly alongside the GP into specific portfolio companies.

- Typical fee: 0 – 1% + reduced carry on the co-invest slice
- Control: medium — deal-by-deal opt-in
- Suited for: experienced LPs seeking fee relief

DIRECT INVESTING

Invest directly in assets without a fund intermediary. Full diligence, structuring, and management done in-house.

- Typical fee: none (internal costs only)
- Control: maximum
- Suited for: largest SWFs, pensions — they compete with GPs

Source: CFA Institute, Curriculum Vol 8, Learning Module 1, pp. 12–18.

Limited partnerships concentrate capital from many LPs and decision-making in a single GP — whose incentives must be carefully aligned through complex compensation terms

LPs commit capital over 10y; GP draws it down as deals close (capital calls). GP runs the business and bears (theoretically) unlimited liability.

ROLES

- GP = Fund Manager
  - Sources, selects, manages, exits deals
  - Receives management fee + carried interest
  - ≥ 2% LP commitment common (skin-in-game)
- LPs = Outside investors
  - Passive, no role in day-to-day decisions
  - Commit capital; small upfront cash outflow
  - Capital calls drawn over 3–5y investment period

TENDERLEDGE FUND VIII — TERM SHEET

- Size: up to USD 750m
- GP commitment: ≥ 2% of aggregate LP commitments
- Term: 10 years from initial closing
- Management fee: 1.5% p.a. on committed capital
- Hurdle rate: hard 10% preferred return
- Performance fee: 20% above the hurdle
- Waterfall: whole-of-fund (European) with clawback
- Side letters: bilateral, override the LPA for one LP

Source: CFA Institute, Curriculum Vol 8, Learning Module 1, Exhibit 3 (Tenderledge term sheet), pp. 19–28.

SECTION 02

Performance & fees

The private-capital J-curve records a predictable sequence — early fees depress returns before capital-deployment gains and, eventually, exit distributions dominate

Because cash flows are lumpy and staged, periodic "TWR" on a PE portfolio is misleading — IRR is the only sensible headline.

1 · COMMITMENT

Years 0–2: Fund is closed, LPs committed, GP identifies deals.

Fees & setup costs accrue IMMEDIATELY; little or no invested capital → return is NEGATIVE.

This is the bottom of the "J".

2 · DEPLOYMENT

Years 2–5: Capital is called and invested; portfolio companies restructure / grow.

Returns begin to compound but management fees still drag. Valuations appraisal-based — often anchored near cost.

3 · DISTRIBUTION

Years 5–10+: Successful exits trigger distributions (trade sale, IPO, SPAC, recap).

Returns accelerate — the "up-stroke of the J". Cash distributions flow back to LPs net of carry.

Source: CFA Institute, Curriculum Vol 8, Learning Module 2, Exhibit 1, pp. 36–44.

IRR captures the timing of cash flows; MOIC captures the multiple — they answer different questions and both are needed

MOIC treats USD 1 returned in year 2 identically to USD 1 returned in year 15. IRR does not. Both matter.

$$\text{IRR:} \quad \Sigma \; CF_t / (1 + IRR)^t = 0 \qquad \text{MOIC} = (\text{Realised} + \text{Unrealised}) / \text{Invested capital}$$

where: Invested capital = paid-in capital less management fees and fund expenses.

KNOWLEDGE CHECK — HIMITSU INVESTS IN ZZZ

Year 0: invest JPY 3.8bn. Year 2: +JPY 1.2bn. Year 3: +JPY 0.2bn. Year 8: sell for JPY 8.5bn.

1. Total invested: 3.8 + 1.2 + 0.2 = JPY 5.2bn
2. MOIC = 8.5 / 5.2 = 1.63×
3. IRR ≈ 20% p.a. (solves NPV = 0 across the staged cash flows)
4. Reading: 1.63× MOIC sounds modest — until you see it was earned in 8 years, giving a ~20% IRR.

MOIC = 1.63× · IRR ≈ 20%. A 2× MOIC over 15 years is a 4.7% IRR — the timing matters.

[1] DPI (Distributions to Paid-In) = Σ cash distributed / Σ paid-in capital — "realised" return. TVPI = (Distributions + NAV) / Paid-in — total including unrealised.
[2] RVPI = NAV / Paid-in. TVPI = DPI + RVPI; as the fund matures, RVPI falls to zero and TVPI converges to DPI.

Source: CFA Institute, Curriculum Vol 8, Learning Module 2, pp. 46–52.

The 2-and-20 fee structure pairs a 2% management fee on assets with 20% carried interest above an 8% hurdle — the industry default

PE typically bases mgmt fee on COMMITTED capital (disincentivises rush-to-deploy). Hedge funds base mgmt fee on AUM.

$$r_GP = \max[\,0\,,\ p \times (r - r_h)\,] \qquad \text{(no catch-up)}$$

where: r = single-period fund return; r_h = hurdle rate; p = GP performance-fee rate (20% is standard).

CFA LM 1 — FUND EARNS 18%, HARD 8% HURDLE, 20% CARRY

Ignoring management fees. Single-period fund rate of return r = 18%. Hard hurdle r_h = 8%. GP carry p = 20%. Compute the GP's share and the LP's share.

1. Excess above hurdle: 18% − 8% = 10%
2. GP share (no catch-up): r_GP = 20% × 10% = 2.0%
3. LP share: 18% − 2.0% = 16.0% → the LP receives the full 8% hurdle PLUS 80% of the 10% excess = 8% + 8% = 16%
4. Sanity check: GP + LP = 2.0% + 16.0% = 18.0% ✓

GP = 2.0% · LP = 16.0%. Under a hard hurdle without catch-up, GP only earns on the excess.

Source: CFA Institute, Curriculum Vol 8, Learning Module 1, Compensation structures, pp. 22–28.

Catch-up clauses accelerate the GP's share above the hurdle; high-water marks and clawbacks limit double-dipping

Same 18% / 8% / 20% fund — GP walks away with 2.0% (no catch-up) or 3.6% (full catch-up). That's a 1.6 pp swing on a single deal.

$$r_GP = \max[\, 0 \,,\; r_cu + p \times (r - r_h - r_cu) \,]$$

(with catch-up)

where: r_cu = catch-up return paid entirely to GP after LP receives hurdle; r_cu such that cumulative split reaches 80/20.

WITH FULL CATCH-UP

LP earns the 8% hurdle first. Then GP receives 100% of next returns until total split = 80/20. Remaining split 80/20.

- LP gets 8% + 6.4% + 0% = 14.4%
- GP gets 0% + 2.0% + 1.6% = 3.6% (vs 2.0% without)

PROTECTIONS AGAINST DOUBLE-DIP

- High-water mark (HWM) — GP earns carry only above fund's previous NAV peak. Prevents paying twice for the same gain.
- Clawback — LPs reclaim carry already paid when later losses bring the fund below the hurdle on aggregate.

Source: CFA Institute, Curriculum Vol 8, Learning Module 1, Exhibits 9–11, pp. 22–28.

SECTION 03

Private capital — PE & private debt

Private equity spans a company's life cycle from idea (VC) to maturity (growth) to turnaround (buyout) — each stage with its own risk, return, and skill set

VC investors underwrite product-market fit. Buyout investors underwrite operational turnaround. Growth equity sits in between.

Stage	Revenues	Cash flow	Business risk	Typical PE form
Seed / pre-seed	None	Negative	Very high	Angel / VC
Start-up	Low to none	Negative	High	Early-stage VC
Growth	Increasing	Increasing	High	Late-stage VC / Growth
Maturity	Positive, predictable	Positive, predictable	Low	LBO / buyout
Decline / turnaround	Declining	Declining	Increasing	Buyout / distressed

EXIT ROUTES DIFFER BY STAGE

VC typically exits via IPO or trade sale to strategic buyer. Buyouts exit via IPO, SPAC, secondary sale (PE-to-PE), or dividend recap. Average hold period: ~5 years — but 6 months to 10+ years in the tails.

Source: CFA Institute, Curriculum Vol 8, Learning Module 3, Exhibits 1–2, pp. 68–72.

A leveraged buyout creates equity value through three levers — cash-flow growth, multiple expansion, and debt paydown — quantified in a textbook IRR build

Figures below are illustrative for training. Real-world LBOs layer in fees, transaction costs, add-on M&A, and management incentive plans.

TEXTBOOK BUYOUT — 5-YEAR HOLD, 60% DEBT, 6× → 7× EBITDA

Acquire target: $EBITDA_0$ = USD 100; 6.0× entry → EV = USD 600. Capital: USD 360 debt (60%) + USD 240 equity. Hold 5 years; EBITDA grows 6% p.a. → $EBITDA_5$ = USD 134. Exit at 7.0× → EV_5 = USD 938. Debt paid down to USD 210.

1. Entry equity: 240; Exit enterprise value: 7.0 × 134 = 938
2. Exit equity = EV − Net debt = 938 − 210 = 728
3. Total equity multiple (MOIC) = 728 / 240 = 3.03×
4. IRR = (3.03)^(1/5) − 1 ≈ 24.8%
5. Decomposition: EBITDA growth +34%, multiple expansion +17%, debt paydown +150 → ≈ 25% IRR over 5y

MOIC ≈ 3.0× · IRR ≈ 25%. Leverage amplifies — a 10% EBITDA miss would drop equity by ~25%.

Source: CFA Institute, Curriculum Vol 8, Learning Module 3, Private Equity Investment Characteristics, pp. 68–78.

Trade sale, IPO, SPAC, recap, secondary, write-off — PE exit routes are not equal and the choice shapes realized returns

Managers pre-plan exit route at entry. Strategic buyer at a synergy premium is still the single most common PE exit.

Exit route	Key advantage	Key drawback
Trade sale	Strategic buyer pays synergy premium; fast execution	Limited universe of strategic buyers; regulatory scrutiny
IPO	Highest potential price; management retention	High cost, long lead time, market volatility, lockup
SPAC	Faster than IPO; fixed valuation; forward guidance	Dilution from warrants; post-merger stockholder overhang
Secondary sale (PE-to-PE)	Liquidity without public-market exposure	Price discovery limited to other sponsors
Recapitalization (dividend recap)	Extract cash without losing control; boosts IRR	Increases leverage on portfolio company
Write-off / liquidation	Closes failed deal, frees team for next project	Signals loss — drags fund performance and track record

Source: CFA Institute, Curriculum Vol 8, Learning Module 3, Exhibit 3, pp. 73–77.

Private debt fills the gap between bank loans and public bonds — four flavours: direct lending, mezzanine, venture debt, distressed

Unitranche and leveraged loans are variants; covenants, seniority, and collateral define risk far more than nominal coupon.

Category	Seniority	Typical borrower	Yield pickup*	Risk profile
Direct lending	Senior, secured	Mid-market corporates	+300–500 bps	Moderate
Mezzanine	Sub to senior, sr to equity	LBOs, recaps, acquisitions	+800–1200 bps	High — with warrants
Venture debt	Senior to equity	Start-ups / early-stage	+500–1000 bps	High — equity-like
Distressed debt	Varies by tranche	Firms at / near bankruptcy	+1000 bps+	Very high — workout risk
Unitranche	Blended senior / sub	Mid-market LBOs	+400–700 bps	Moderate-to-high

***YIELD PICKUP OVER COMPARABLE PUBLIC DEBT**

Indicative ranges from broker research. Rates quoted off a reference rate (SOFR / MRR) — coupon floats with policy-rate moves. The illiquidity premium is real but hard to measure because appraisal-based marks smooth observed volatility.

[1] Private-debt market grew after the 2008 crisis as banks retrenched under tightened regulation. Private-debt funds filled the gap, charging an illiquidity premium.

Source: CFA Institute, Curriculum Vol 8, Learning Module 3, pp. 79–84.

Risk and return rise smoothly as you move from senior infrastructure debt up the capital stack to private equity co-investments — each rung adds a specific risk premium

Empirical correlations of private capital with public indexes range 0.63–0.86 (Preqin vs MSCI World / S&P 500) — meaningfully positive.

Rung	Position	Risk premium paid for…
1 · Infrastructure debt	Most senior real-asset debt	Illiquidity; long duration
2 · Senior real-estate debt	First-lien mortgages; IG CMBS	Illiquidity; collateral diligence
3 · Senior direct lending	Senior secured corporate loans	Credit risk of non-public borrowers
4 · Unitranche debt	Blended senior / subordinated	Credit + subordination risk
5 · Mezzanine debt	Sub to senior, sr to equity	Subordination + embedded equity upside
6 · Private equity (buyout / growth)	Common equity in private company	Leverage, illiquidity, concentration
7 · PE co-investments	Concentrated direct equity stakes	All of PE + deal-specific concentration

Source: CFA Institute, Curriculum Vol 8, Learning Module 3, Exhibit 6, pp. 85–87.

SECTION 04

Real estate & infrastructure

Real estate returns run a spectrum from bond-like core to equity-like opportunistic — leverage and development risk drive the escalation

The deeper you go up the ladder, the more of total return comes from price appreciation rather than rental income.

Strategy	Character	Example deal
Senior debt	Bond-like; first mortgages, investment-grade CMBS	IG CMBS pool
Core	Stable income producing; long leases; many lessors; low leverage	Trophy CBD office
Core-plus	Below-IG CMBS, tenant repositioning, stabilising cash flow	Minor refurbishment
Value-add	Re-leasing, redevelopment, property repositioning	B-grade office retrofit
Opportunistic	Development, distressed, mezzanine RE debt, major redevelopment	Ground-up development

WHAT DRIVES THE ESCALATION

Leverage (LTV rises from ~50% in core to 70%+ in value-add/opportunistic), vacancy risk (stabilised vs speculative leasing), construction risk (zoning, cost overruns, permitting), and market risk (cyclical cap-rate compression / expansion).

Source: CFA Institute, Curriculum Vol 8, Learning Module 4, Exhibit 4, pp. 102–106.

Direct real estate is valued off a cap rate on stabilised NOI; public REITs trade on an FFO multiple — the two numbers should reconcile but rarely do in real time

NAV-to-market premium/discount persists for public REITs — daily pricing vs annual appraisals creates a persistent gap.

DIRECT · CAP RATE ON NOI

Office building: annual rents USD 8m, opex USD 2m, vacancy factor already included.

1. NOI = Rents − Opex = 8 − 2 = USD 6m
2. Market cap rate for comparable CBD office: 5.5%
3. Value = NOI / cap rate
4. = 6 / 0.055 = USD 109m

Direct RE value ≈ USD 109m

REIT · FFO MULTIPLE

REIT: Net income USD 1.60/sh; RE depreciation USD 0.80/sh; no property sales this year.

1. FFO/sh = 1.60 + 0.80 = USD 2.40
2. Peer-group FFO multiple: 15×
3. Implied share price = 15 × 2.40
4. = USD 36/sh

REIT implied price ≈ USD 36/sh

[1] FFO (Funds From Operations) = Net income + Real-estate depreciation − Gains on property sales. A REIT-specific GAAP-adjustment because RE depreciation is largely non-economic.

Source: CFA Institute, Curriculum Vol 8, Learning Module 4, REIT valuation and FFO, pp. 100–106.

REITs remove double taxation by distributing 90-100% of taxable rental income — the price of that gift is limited retained capital for growth

REIT vs direct RE: daily liquidity but higher correlation with public equity; lower minimum ticket but reduced control over property decisions.

EQUITY REITs

Own & operate income-producing property. Maximise occupancy × rent minus opex.

- 90–100% payout of taxable income
- Primary return: dividend + appreciation
- Valued on FFO / AFFO multiples

MORTGAGE REITs

Own real-estate debt instead of real estate itself.

- Portfolio of MBS and whole mortgages
- Highly leveraged; NIM-driven return
- Sensitive to interest-rate moves and credit spreads

HYBRID REITs

Mix equity and debt real-estate exposure in one vehicle.

- Lower sector-specific risk
- Return driven by mix of rental + interest income
- Less common than pure-equity or pure-mortgage REITs

Source: CFA Institute, Curriculum Vol 8, Learning Module 4, pp. 98–106.

Infrastructure is real-asset cousin to real estate but distinguished by public-use purpose, concession contracts, and long-duration availability-based cash flows

Core infra cash flows look bond-like — regulated utility rates, availability payments, long-tenor tolls — attractive for LDI / pension matching.

Category	Examples	Cash-flow mechanism
Economic — Transportation	Toll roads, bridges, airports, ports, rail	Usage-based (traffic, tolls)
Economic — ICT	Telecom towers, fibre, data centres	Long-term leases to carriers
Economic — Utilities / energy	Power grid, gas pipelines, water, waste, LNG terminals, renewables	Regulated tariffs / take-or-pay
Social — Education & health	Schools, hospitals, clinics, correctional facilities	Availability payments from gov't
Social — Gov't buildings	Municipal buildings, social housing	Lease from gov't under PPP

FINANCING STRUCTURE

Public–private partnership (PPP) under a long-term concession agreement — private investor builds, operates, and maintains; government retains underlying right and often buys off at the end. Typical concession tenor: 20–40 years.

Source: CFA Institute, Curriculum Vol 8, Learning Module 4, pp. 107–116.

The greenfield-to-brownfield spectrum maps infrastructure risk: build-operate-transfer projects start with negative cash flows, stabilise through operations, and exit to long-duration investors

Infra J-curve resembles PE: build phase is cash-negative; operate phase is cash-positive; transfer phase realises terminal value.

GREENFIELD

Build new asset from scratch.

- Highest risk: construction, permitting, demand
- BOT life cycle: Build → Operate → Transfer
- Expected return highest
- Typical investor: dev specialists + strategic partners

BROWNFIELD

Expand / upgrade existing asset, often via privatisation or sale-leaseback.

- Operating history available → lower risk
- Immediate cash flows
- Moderate expected return
- Investor: strategic + financial mix

SECONDARY-STAGE

Fully operational asset, no further material capex.

- Lowest risk — pure cash-flow asset
- Stable, predictable distributions
- Lowest expected return
- Investor: pensions, insurers, SWFs seeking duration

Source: CFA Institute, Curriculum Vol 8, Learning Module 4, Exhibit 7, pp. 109–112.

SECTION 05

Natural resources

Raw land, farmland, and timberland share real-estate DNA but derive their value from the soil and the biology — not the buildings

Timberland acts as both factory AND warehouse — you can delay harvest when lumber prices are weak. Farmland cannot.

Attribute	Raw land	Farmland	Timberland
Return drivers	Land price	Harvest qty · commodity prices · land	Biological growth · lumber · land
Source of revenue	Lease income (limited)	Crop sales + lease revenue	Timber sales + lease revenue
Main risks	Alternative-use optionality	Weather, climate, disease	Weather, disease, wildfire
Ownership forms	Direct / partnership	Direct / partnership / REIT	Direct / partnership / REIT / TIMO
Typical owner	Mostly individuals	Family + inst'l	Mostly institutional

TIMBERLAND = FACTORY + WAREHOUSE

Unique to timberland: trees keep growing when you don't harvest (biological growth continues). Farmland: harvest-or-lose-the-crop (no warehouse). This asymmetry is a core timberland selling point — it adds optionality on lumber prices.

Source: CFA Institute, Curriculum Vol 8, Learning Module 5, Exhibit 1, pp. 124–128.

Commodity total return equals spot return plus roll yield plus collateral yield — roll yield is where the shape of the forward curve enters portfolio P&L

Most commodity exposure is via futures. When the front contract rolls to the next month, the slope of the curve determines whether you bleed or gain.

$$F_0(T) = S_0 \times e^{(r + c - i) \times T}$$

$$\text{Total return} = \text{Spot return} + \text{Roll yield} + \text{Collateral yield}$$

where: c = cost of carry (storage, insurance, transport); i = convenience yield (non-cash benefit of holding the physical); r = risk-free rate.

BACKWARDATION · ENHANCES RETURN

Forward curve SLOPES DOWN (future prices < spot).

- Convenience yield > cost of carry
- Typical when inventories are LOW
- Long-only investor ROLLING forward BUYS CHEAP, sells rich → positive roll yield

CONTANGO · DRAG ON RETURN

Forward curve SLOPES UP (future prices > spot).

- Cost of carry > convenience yield
- Typical when inventories are HIGH
- Example: crude oil April 2020 flipped to deep contango as COVID crushed demand — long-only investors bled roll yield

Source: CFA Institute, Curriculum Vol 8, Learning Module 5, pp. 130–140.

Commodities have historically outperformed stocks and bonds during rising-inflation regimes — farmland and timberland add lower-correlation diversification

Data 1992–2022. Commodities annualised 7.81% at 24.4% vol; rolling regimes matter as much as the headline return.

Metric	Global Stocks	Global Bonds	Commodities	Farmland	Timberland
Annualised return 1992–2022	6.89%	4.39%	7.81%	10.95%	8.69%
Annualised standard deviation	16.8%	6.1%	24.4%	5.9%	6.8%
Return — HIGH inflation years	+9.4%	+5.7%	+22.9%	+12.8%	+10.4%
Return — LOW inflation years	+5.4%	+4.2%	−9.3%	+9.9%	+5.7%
Correlation w/ global stocks	1.00	—	0.41	0.12	0.02
TAKEAWAY			**Inflation hedge**	**Stable**	**Lowest ρ**

Source: CFA Institute, Curriculum Vol 8, Learning Module 5, Exhibits 6 & 8, pp. 136–140.

SECTION 06

Hedge funds

Hedge funds are defined by investment approach — leverage, shorting, derivatives — not by the underlying assets, which makes the strategy taxonomy the right way to classify them

HFR, MSCI, and Preqin all converge on roughly these five buckets. Multi-strategy / multi-manager funds combine several of them under one GP.

Strategy family	Sub-strategies	Typical characteristic
Equity hedge	Fundamental L/S, market neutral, short biased, fund'l growth / value	Bottom-up stock selection; variable beta
Event-driven	Merger arb, distressed, special situations, activist	Corporate-catalyst-driven
Relative value	Convertible arb, fixed-income arb, multi-strategy	Market-neutral by design; levered spreads
Opportunistic	Global macro, managed futures / CTAs	Top-down; directional on macro views
Multi-manager / FoHF	Pod shops (Millennium, Citadel), fund of hedge funds	Diversification across PMs & strategies

Source: CFA Institute, Curriculum Vol 8, Learning Module 6, Exhibit 1, pp. 150–157.

Equity long-short is the original hedge fund strategy; market-neutral, short-bias, and fundamental growth/value are variants of the same long-plus-short core

The market beta of each variant is chosen deliberately — from close to +1 (fund'l growth) through 0 (market neutral) to near −1 (short biased).

Variant	Net exposure	Typical beta	Primary alpha source	Downside
Fundamental L/S (long-biased)	+30% to +60%	+0.4 to +0.7	Long winners vs short losers within same sector	—
Fundamental growth	Long-biased	+0.6 to +1.0	Long high-growth; short declining business models	—
Fundamental value	Long-biased	+0.4 to +0.8	Long undervalued turnarounds; short overpriced	—
Market neutral	≈ 0%	near 0	Pure security selection; stable low single-digit	—
[illegible]	[illegible]	[illegible]	Forensic analysis;	

CORRELATION EVIDENCE (LM6 EXAMPLE 1)

Over 10y monthly data: fundamental value / growth correlate 0.80 / 0.67 with the market; market-neutral is 0.22; short-bias is −0.72. Market-neutral delivers stable sub-1% monthly vol but leverage is typical to amplify the low single-digit edge.

Source: CFA Institute, Curriculum Vol 8, Learning Module 6, pp. 152–154.

Event-driven and relative value strategies profit from specific corporate catalysts and temporary pricing dislocations — both use leverage to amplify modest spreads

Both share a "spread compression" thesis — and both can get blown up when the catalyst fails (deal break, default, liquidity crunch).

EVENT-DRIVEN

- Merger arbitrage — long target, short acquirer; profit as announced spread narrows to close. Risk: deal break.
- Distressed / restructuring — buy bonds at deep discount pre-bankruptcy; "fulcrum security" for reorg equity.
- Special situations — spinoffs, recaps, rescue finance, catalyst-driven equity.
- Activist — take board seat, push for divestitures / buybacks / strategy change.

RELATIVE VALUE

- Convertible-bond arbitrage — long convert, short delta-equivalent stock; profit from mispricing of embedded option.
- Fixed-income arbitrage — sovereign curve, yield-curve trades, credit spread vs treasuries, on-the-run vs off-the-run.
- RE / asset-backed RV — mortgage-backed, high-yield spread trades.
- Multi-strategy — deploy capital across all of the above, reallocate dynamically.

Source: CFA Institute, Curriculum Vol 8, Learning Module 6, pp. 154–156.

Hedge funds typically charge 2-and-20 with a high-water mark and impose lockup / redemption gates — features that reshape investor liquidity and fee math

"1-or-30" is emerging: manager earns greater of 1% mgmt fee or 30% of alpha vs benchmark. Realigns incentive around outperformance, not AUM.

FEES

Standard 2-and-20:
- 2% mgmt on AUM (not committed)
- 20% of profit above HWM

Emerging 1-or-30:
- Greater of 1% or 30% × alpha

FoHF: extra +1% mgmt and +10% incentive layered on top.

STRUCTURES

- Master-feeder — onshore & offshore feeders into one master fund; tax efficiency.
- Separately managed account (SMA) — assets in investor's name; investor retains custody and legal title.
- Fund of one — single LP.

LIQUIDITY RESTRICTIONS

- Lockup — minimum hold period (1–3y typical).
- Soft lockup — early redemption allowed with 1–5% fee payable to the fund.
- Redemption notice — 30–90 days.
- Gate — fund-wide cap on % of NAV that can be redeemed in any quarter.

Source: CFA Institute, Curriculum Vol 8, Learning Module 6, pp. 158–166.

SECTION 07

Digital assets

Distributed ledger technology turns a trusted database into a trustless network — every full node holds a verified copy; consensus replaces the central intermediary

Trade-off at the heart of DLT: decentralisation costs energy and throughput. Permissioned networks trade some decentralisation for speed.

PROOF OF WORK (PoW)

Miners solve a computationally expensive puzzle to propose the next block.

- Energy-intensive (Bitcoin ≈ small country's draw)
- Secured by economic cost of attacking
- 51% attack = control most computing power
- Used by Bitcoin, legacy Ethereum (pre-Merge)

PROOF OF STAKE (PoS)

Validators lock ("stake") capital to vouch for a block's validity.

- Far lower energy cost
- Secured by economic cost of losing the stake
- Ethereum moved PoW → PoS in Sept 2022 (The Merge)
- Used by Ethereum, Solana, Cardano, etc.

PERMISSIONED vs PERMISSIONLESS

Permissionless (Bitcoin, Ethereum) = open to anyone, governance decentralised. Permissioned (many enterprise chains) = gated membership, faster & cheaper but requires trust in gatekeeper. Governance and throughput trade off against open participation.

Source: CFA Institute, Curriculum Vol 8, Learning Module 7, pp. 181–184.

Digital assets split into cryptocurrencies and tokens — with distinct investment characteristics and regulatory ambiguity across both

Bitcoin + Ether ≈ 80% of crypto market cap (as of July 2022). Everything else is the long tail.

Type	Sub-type	Use case
Cryptocurrencies	Bitcoin	Medium of exchange / store of value
Cryptocurrencies	Altcoins (Ether, Solana, …)	Programmable smart contracts
Cryptocurrencies	Stablecoins (USDC, Tether)	Peg to fiat — settlement rail
Cryptocurrencies	CBDCs (pilot: eCNY, digital euro)	Tokenised fiat issued by central bank
Tokens	Security tokens (ICO)	Digitised ownership of a security
Tokens	Utility tokens	Pay for services on a network
Tokens	Governance tokens	Vote on protocol changes
Tokens	NFTs	Unique authenticated digital object

INVESTMENT FORMS

Direct (wallets on centralised / decentralised exchanges) · Indirect (crypto futures on CME, spot Bitcoin ETFs, listed miners, hedge funds).

Source: CFA Institute, Curriculum Vol 8, Learning Module 7, Exhibit 4 & pp. 185–206.

Bitcoin and Ether have offered high returns, high volatility, and low correlation — but tokenisation, custody, and regulatory risks are existential, not academic

Digital-asset risk is not a list of price drawdowns — it is a list of ways your position can go to zero independent of price.

RISK 1 · CUSTODY

- Centralised exchanges hold your keys — hacks & fraud (Mt. Gox, FTX) wipe out balances.
- Self-custody shifts the risk to you — lost seed phrase = irrecoverable loss.
- Institutional qualified custodians (Coinbase, Fidelity Digital) emerging but nascent.

RISK 2 · REGULATORY

- Jurisdictional patchwork: legal in some countries, banned in others.
- SEC / CFTC turf war on "security" vs "commodity" status of specific tokens.
- Tax treatment evolving fast — rules from 2020 already outdated.
- Use can be criminal in several jurisdictions.

RISK 3 · VALUATION & PEG

- No cash flows → no fundamental anchor. Value is entirely expectation-driven.
- Algorithmic stablecoins can de-peg fast: Terra/LUNA → USD 40bn market cap to ~0 in one week (May 2022).
- Even collateralised stablecoins (Tether) face reserve-quality scrutiny.

[1] Curriculum text characterises Bitcoin as "high return, high risk, low correlation" — but warns that correlation with risk assets has risen materially post-2022.

Source: CFA Institute, Curriculum Vol 8, Learning Module 7, pp. 188–206.

KEY TAKEAWAYS

Deck 19 — Alternative Investments

- Alts are defined by what they are NOT — three buckets (private capital, real assets, hedge funds) share illiquidity, specialised knowledge, appraisal-based valuations, and performance-based fees.
- J-curve + IRR is the correct performance lens for long-lived alts; MOIC, DPI, TVPI each answer a different question and should be reported together.
- 2-and-20 with 8% hurdle: on a single-period 18% return, GP earns 2.0% (no catch-up) or 3.6% (full catch-up). Clawbacks and HWMs prevent double-dipping.
- PE spans VC → growth → buyout across the life cycle; LBO value creation = cash-flow growth + multiple expansion + debt paydown — a 5y textbook deal clears ~25% IRR.
- Real-estate risk ladder runs from senior debt / core (bond-like) to opportunistic (equity-like); a building at 5.5% cap on USD 6m NOI = USD 109m; its REIT trades on FFO multiples.
- Infrastructure cash flows are bond-like under concession contracts; BOT life cycle separates greenfield (highest risk) from brownfield / secondary (lower risk, lower return).
- Commodities beat stocks & bonds in high-inflation regimes but lose in low-inflation regimes; roll yield (contango vs backwardation) dominates spot return in futures-based strategies.
- Hedge funds classified by STRATEGY, not asset class: equity L/S, event-driven, relative value, macro/CTA, multi-manager. Digital assets add DLT-specific risks — custody, regulation, peg — on top of price volatility.